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Thick AlN Templates By MOCVD for the Thermal Management of III-N Electronics

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Abstract

Ultra-wide bandgap semiconductors have attracted much interest over the last decade. These materials have strong chemical bonding that led to high temperature tolerance, high voltage and power handling capabilities as described by the Baliga figure of merit. Al-rich III-nitride materials are expected to have the best performance over the industry standard Si, and newer options such as SiC and GaN. Continuous improvements dictate the miniaturization of electronic devices, leading to increasingly higher power densities in smaller footprints, leading to heating that causes system failure. Thus, the thermal management is the limiting task for continuous performance scaling. In past we have reported on deep ultraviolet light emitting diodes (LEDs) and $Al_xGa_{1-x}N$ ($x>0.4$) channel Heterojunction Field-Effect Transistors (HFETs) on 2-3 μm thick MOCVD grown high quality AlN/sapphire templates. For template thicknesses over 5 μm either sapphire or AlN patterning is needed to avoid stress related cracking. In this research, we for the first-time report on the direct MOCVD growth and characterization of crack-free, low-impurity AlN templates in excess of 16 μm on basal plane sapphire substrates without doing any external processing. Increasing the thickness improves thermal management and enables an easier sapphire substrate's laser liftoff for increased DUV light extraction. It would also be used as a heat spreader in power electronics to minimize the operating temperature to maximize power output to industrial loads such as electric vehicles, and large turbines in manufacturing and transportation.

All growths were carried out in custom MOCVD reactors with a fast-metalorganic switching manifold using a modified epitaxy procedure in which the Al- and N-precursors (Trimethyl Aluminum and NH_3) are pulsed for increased surface mobility. At the initial pulsed growth stage, the growth-temperature and precursor-flow rates are adjusted to yield air-pocketed (voids) rough AlN layers. Most of these high defect materials are located near the AlN/sapphire interface. As the AlN thickness increases in the subsequent growths, the layers turned out smooth and the number of defects is significantly reduced

due to dislocation bending/annihilation and cracking is avoided due to strain relief from the voids at the interface. All template growths were carried out at temperatures ~ 1200 - 1300 °C and 40 torr.

From X-ray diffraction (XRD), Reciprocal space map (RSM), and Transmission Electron Microscopy (TEM) analysis of these thick AlN templates, we established that they were fully relaxed. Cross-section TEM data clearly shows the presence of voids at ~ 0.4 – 0.9 μm from the AlN/sapphire interface. There is a high density of dislocations present near the void region. Above this region, at least one order of magnitude lower dislocations is observed at the AlN smooth layer region. From the room-temperature Cathodoluminescence (CL) data we see an increase in the band edge emission and a decrease in the defect related long-wavelength signal with increasing AlN template thicknesses. The CL spectra were measured in the edge emission pump geometry on cleaved bars due to the transverse-magnetic polarized nature of the band-edge emission. The monochromatic CL imaging data also confirmed a highly defective region at the AlN/sapphire interface which is also the origin of the long wavelength (below bandgap) CL emission. The Atomic Force Microscopy (AFM) surface scans of the templates showed the surface roughness ~ 0.15 nm to 0.25 nm (for a 5 μm x 5 μm scan). The (102) off axis X-ray linewidths for all these samples ranged from 280 to 330 arc-secs. XRD and Raman experiments confirmed a residual compressive strain in AlN templates compared to the bulk sample. The measured stress ranges from -0.6 GPa to -1.2 GPa for all the templates, reasonable agreement with previous reports. We have found no obvious thickness dependence of the stresses measured from both XRD and Raman, an indication of preserving similar stresses in thick and thin templates.

The thermal conductivity of our high-quality 16 μm thick template and the bulk AlN are measured at different temperatures by Time-domain Thermoreflectance (TDTR) technique. The thermal conductivity of the thick AlN at room temperature is 320 $Wm^{-1}K^{-1}$, or ~10% higher than that of the bulk AlN (~285 W/m-K). At low temperatures (120 K), it exceeds that

of bulk AlN by more than 40%. In summary, these templates showed thermal conductivity as high or better than high-cost bulk AlN substrates, demonstrating for the first time that sapphire-based power electronics could be effectively managed thermally.

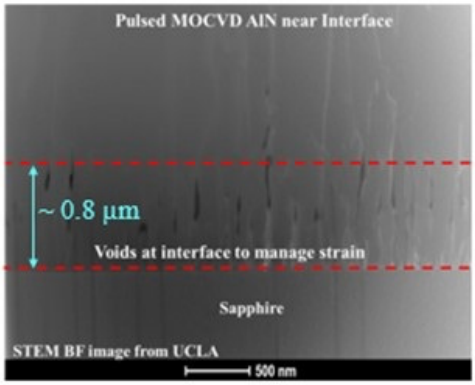


Figure 1. Cross-section TEM image at the bottom 2 µm of a 3 µm AlN template showing voids at the interface.

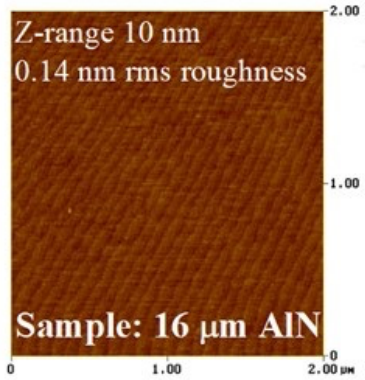


Figure 2. smooth surface morphology of a 16 µm template by Atomic Force Microscopy (AFM).

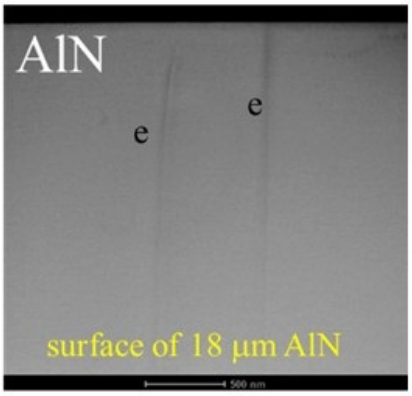


Figure 3. (0002) two-beam condition cross-sectional STEM BF image near the surface (top 3.3 µm) of an 18 µm thick template showing reduced dislocations.

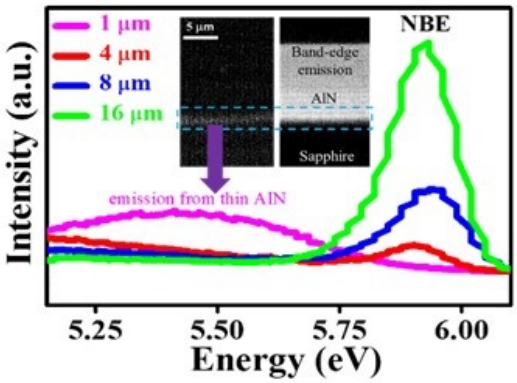


Figure 4. CL spectra of different thick templates with cross-section CL image of a 16 µm template at two different wavelengths.

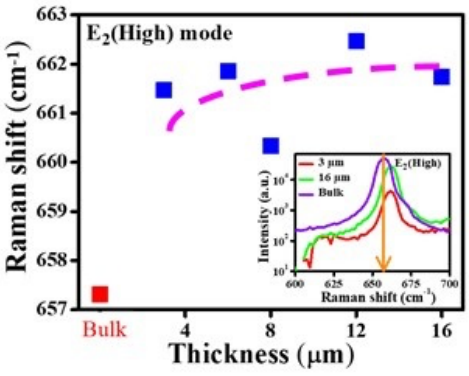


Figure 5. Raman shift and (inset) Raman spectra of different thick AlN templates with the bulk one. The error bars are elongated 5X for better visualization.

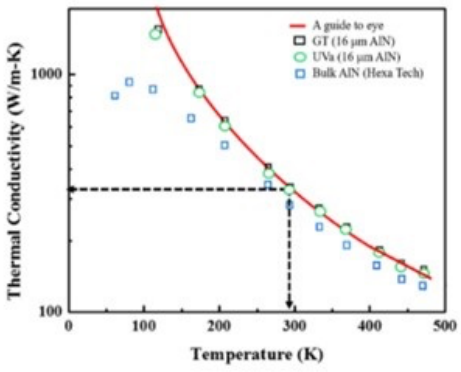


Figure 6. Time-Domain Thermoreflectance (TDTR) data for the 16 µm thick template in comparison with the bulk AlN from HexaTech.

Figure 1

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